

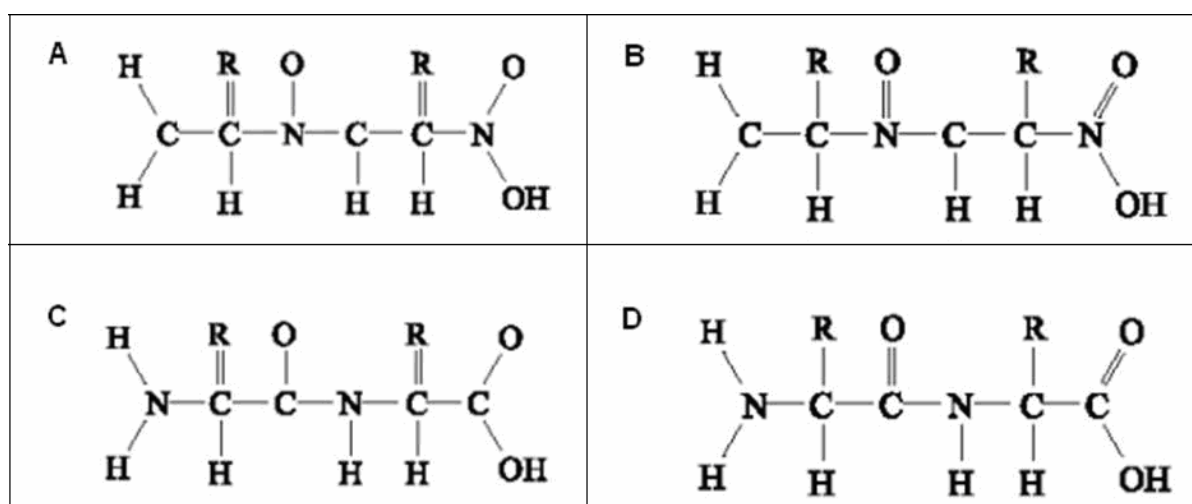
1. Phospholipids, RNA and DNA are present in all or some of the organelles. Choose the most appropriate option.

- A Ribosomes are the only organelles that have all the three compounds.
- B Both smooth endoplasmic reticulum and mitochondria have only phospholipids.
- C Both ribosomes and mitochondria have phospholipids and DNA.
- D Mitochondria are the only organelles that have all the three compounds.**

2. Which organelles have a transport function?

- A endoplasmic reticulum and Golgi apparatus**
- B Golgi apparatus and ribosome
- C mitochondrion and endoplasmic reticulum
- D ribosome and mitochondrion

3. Which is the structure of a dipeptide? **D**



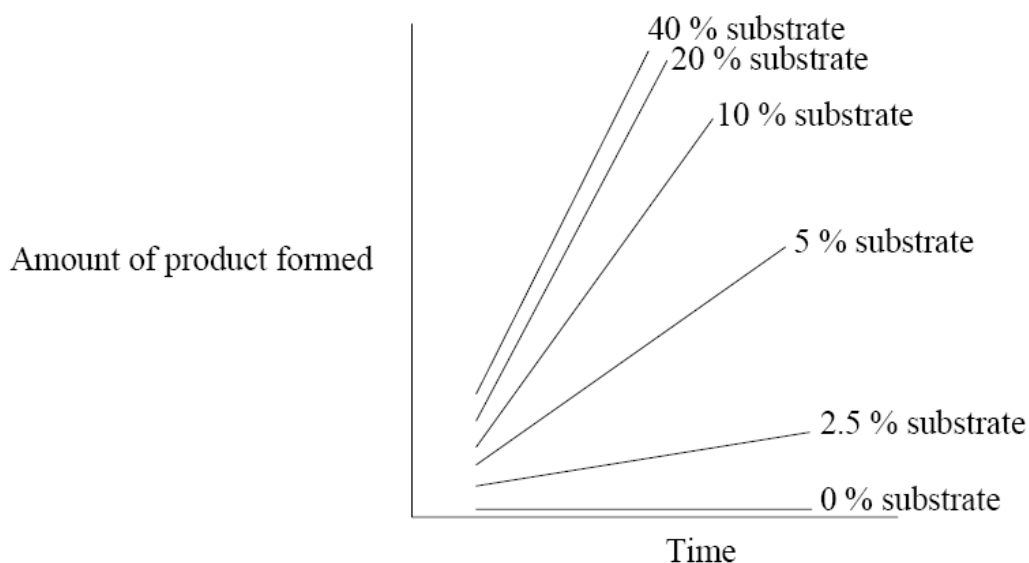
4. Three pure substances are analysed and the elements that each contains are listed in the table below:

	C	H	O	N	S	P
E	+	+	+	-	-	-
F	+	+	+	+	-	-
G	+	+	+	+	+	-

Which combination could be correct?

	E	F	G
A	adenine	carbohydrate	fat
B	fat	adenine	amino acid
C	amino acid	fat	carbohydrate
D	fat	carbohydrate	amino acid

5. The graph below shows the effect of changing the substrate concentration on an enzyme controlled reaction.



What is the correct interpretation of these data?

- A** The rate of reaction increases continuously with increase in substrate concentration.
- B** The rate of reaction decreases continuously with increase in substrate concentration.
- C** The rate of reaction increases up to a point and then remains constant.
- D** The rate of reaction is not affected by any change in the substrate concentration.

6. The following events occur during the mitotic cell cycle.

- 1 division of centromeres
- 2 duplication of centrioles
- 3 duplication of chromosomes
- 4 overlapping of non-kinetochore spindle fibres

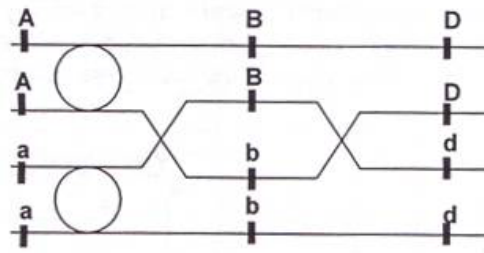
Which two events ensure that the daughter cells are genetically identical to each other and to the parent cell?

- A** 1 and 2
- B** 1 and 3
- C** 2 and 3
- D** 2 and 4

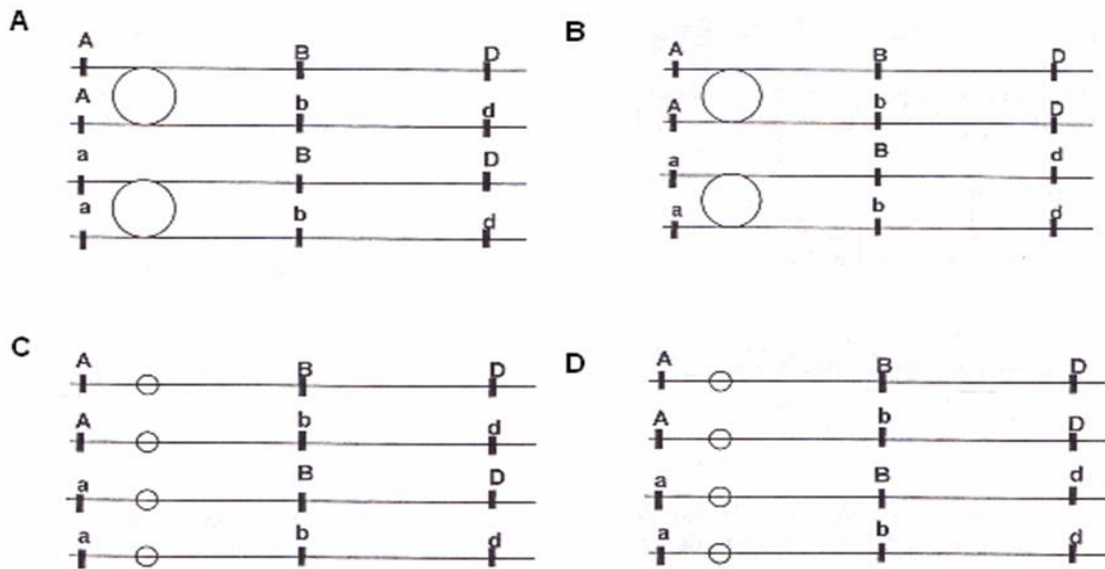
7. Colchicine is a chemical substance that prevents the formation of microtubules. Which phase of mitosis would be prevented if the dividing cells were treated with colchicine.

- A** disintegration of the nuclear membrane
- B** replication of DNA
- C** separation of genetically identical chromosomes (chromatids)
- D** supercoiling of chromosomes

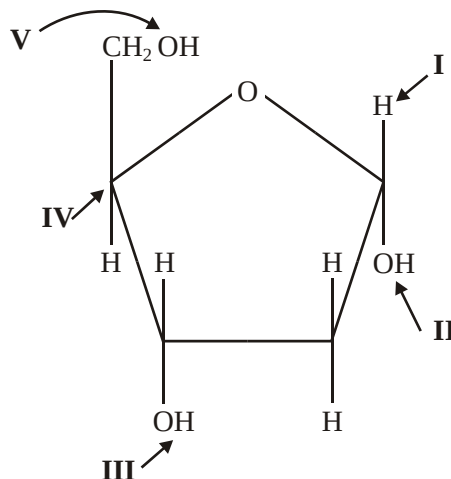
8.



The diagram above shows a bivalent. Which of the following correctly represents the final products at the end of Meiosis II? **D**

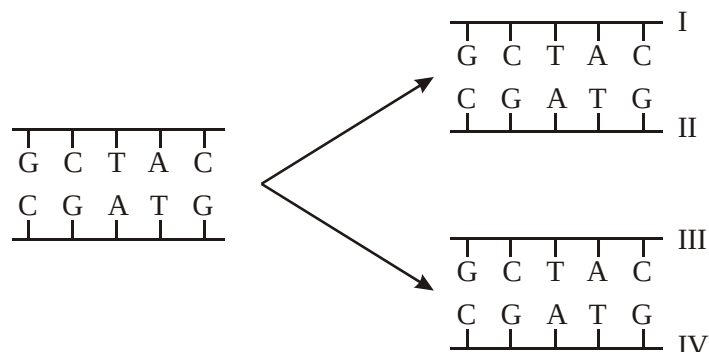


9. To which parts of the deoxyribose molecule do phosphates bind in DNA?



- A I and V
- B III and IV
- C II and III
- D III and V**

10. The diagram below shows a short section of DNA molecule before and after replication. If the nucleotides used to replicate the DNA were radioactive, which strands in the replicated molecules would be radioactive?



- A** II and III only
- B** I and III only
- C** I and II only
- D** I, II, III and IV

11. The steps involved in DNA replication and protein production are summarised below.

Step A: DNA is copied and each new cell gets a full copy.	DNA replication
Step B: Information is copied from DNA and taken to cytoplasm.	Protein synthesis
Step C: Ribosome reads information and assembles protein.	
Step D: Protein formation is completed.	

In which step would a mutation lead to the formation of a new allele?

- A** Step A
- B** Step B
- C** Step C
- D** Step D

12. An mRNA codon for the amino acid alanine is GCC.
How many alanine molecules are present in the polypeptide, containing eight amino acids, coded for by the following DNA template?

DNA template: 3' -CCTACTCGGCCCGGGCCCATGCCAAT- 5'

- A** 0
- B** 2
- C** 1
- D** 3

13. Amino acids complex with transfer RNA through

- A an aminoacyl link with cytosine residue at the 5' end of the chain
- B a phosphodiester bond
- C hydrogen bond
- D an aminoacyl link with a 3' terminal adenosine residue

14. Which one of the following is most likely to occur?

- A When F^+ crosses with F^- , the recipient becomes F^+ , the donor becomes F^- .
- B When F^+ crosses with F^- , the recipient remains F^- , the donor remains F^+ .
- C When Hfr crosses with F^- , the recipient remains F^- , the donor remains Hfr.
- D When Hfr crosses with F^- , the recipient becomes Hfr, the donor remains Hfr.

15. Cells of an *E. coli* strain that are *trp- lac Z- met+* were mixed with cells of an *E. coli* strain that are *trp+ lac Z+ met-* and cultured for several hours. Then cells were removed, washed, and transferred to minimal media containing lactose as the only sugar source. A few cells were able to grow on minimal media with lactose, and formed colonies.

How did these few cells become *trp+ lac Z+ met+*?

- A transformation
- B transduction
- C mutation
- D transposition

16. An *E. coli* strain that is *lac Z-* is conjugated with *E. coli* cells carrying an F plasmid with the *lac P Oc lac Z+* DNA sequence on the episome.

lac P is the lactose promoter. *Oc* is a mutation of the lactose operator that is no longer able to bind the *lac I* gene product which codes for the lac repressor.

How would the expression of the *lac Z+* be regulated in the resulting cells that are diploid for the lactose regulatory region and the *lac Z* gene?

- A normal regulation of lactose metabolism
- B constitutive expression of *lac Z+*
- C inability to synthesize the *lac Z+* gene product, β -galactosidase
- D *lac Z+* gene is inducible, but unable to be repressed by high glucose

17. An *E. coli* cell is grown in the presence of high amounts of glucose. Which of the following is true?

- A The cell will utilize lactose as a carbon source exclusively.
- B The level of cyclic-AMP in the cell will be low.
- C The level of cyclic-AMP in the cell will be high.
- D Transcription of mRNA from the lac operon will be high.

18. In an experiment to investigate the function(s) of capping and polyadenylation, scientists introduced firefly luciferase mRNA, with and without a cap, and with and without poly(A), into tobacco cells. The amount of remaining luciferase mRNA and the luciferase activity were measured experimentally. Luciferase is a protein product that is easy to detect because of the light it generates in the presence of appropriate substances (luciferin and ATP). Luciferase activity would therefore be a reflection of the amount of proteins produced (translation efficiency).

Table below illustrates the effects of capping and polyadenylation on the life-span of the luciferase mRNA and its protein products.

mRNA	Luciferase mRNA half life (min)	Luciferase activity (units)
<u>Uncapped</u>		
Poly(A) tail absent	31	2941
Poly(A) tail present	44	4480
<u>Capped</u>		
Poly(A) tail absent	53	62595
Poly(A) tail present	100	1331917

Half life refers to the time taken for the amount of mRNA to be degraded to half the original amount.

Which of the following is **true**?

- A Capping and polyadenylation have the same effect on translational efficiency.
- B In capped mRNA, presence of poly(A) results in an increase in luciferase activity of 10 times.
- C Polyadenylation protects against mRNA degradation in mRNA more than capping.
- D In polyadenylated mRNA, capping the mRNA results in an increase in luciferase activity of 297 times.

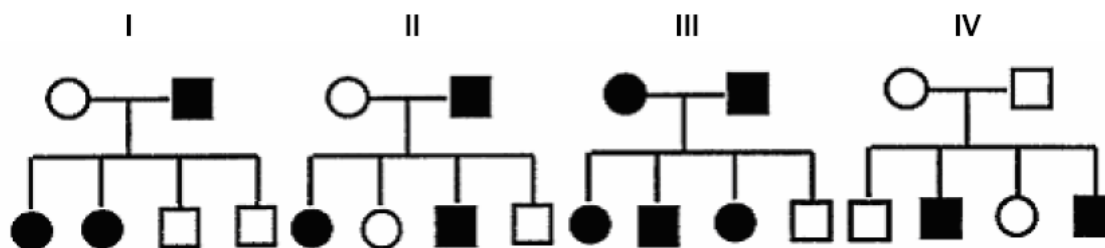
19. Normal cells exhibit density-dependent inhibition when _____

- A More and more cells enter the S phase of the cell cycle.
- B the amount of required growth factors and nutrients per cell becomes insufficient to allow for cell division.
- C the protein kinases they produce begin to compete with each other, such that the proteins produced by one cell essentially cancel those produced by its neighbour.
- D they begin to squeeze against each other, restricting their size and ability to produce transcription factors.

20. How many different genotypes can be found in the offspring of the cross **BbCc** x **BbCc**?

- A 2
- B 4
- C 9
- D 16

21. The figure below shows 4 pedigrees. A shaded circle represents an affected female while a shaded square represents an affected male. Which pedigree(s) represent(s) an inheritance of dominant sex-linked trait?



- A I only
B I and III
 C II and III
 D III and IV

22. In the Sailfin Molly fish, gold colour (g) is recessive to normal colour (G). When a gold coloured fish was crossed with a normal colour fish, 55 of the offspring were normal colour and 45 were gold colour.

One hypothesis is that the normal parent was heterozygous.

The Chi-squared value is 1. The table below shows the probability values.

Degrees of freedom	Probability			
	0.99	0.950	0.05	0.01
1	0.000	0.004	3.84	6.64
2	0.020	0.103	5.99	9.21
3	0.115	0.352	7.82	11.35

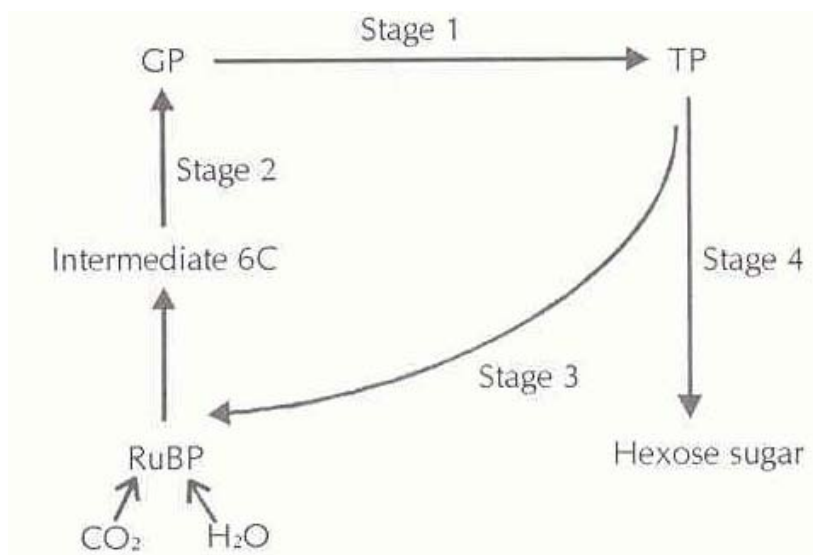
Which of the following is the correct response?

- A Do not reject the hypothesis because the Chi-squared value is less than 3.84.**
 B Reject the hypothesis because the probability is less than 0.05.
 C Do not reject the hypothesis because the Chi-squared value is less than 5.99.
 D Reject the hypothesis because there is not enough evidence.

23. Pigments are extracted from the leaves of a green plant. White light is then passed through the solution of pigments. What effect do the leaf pigments have on the white light?

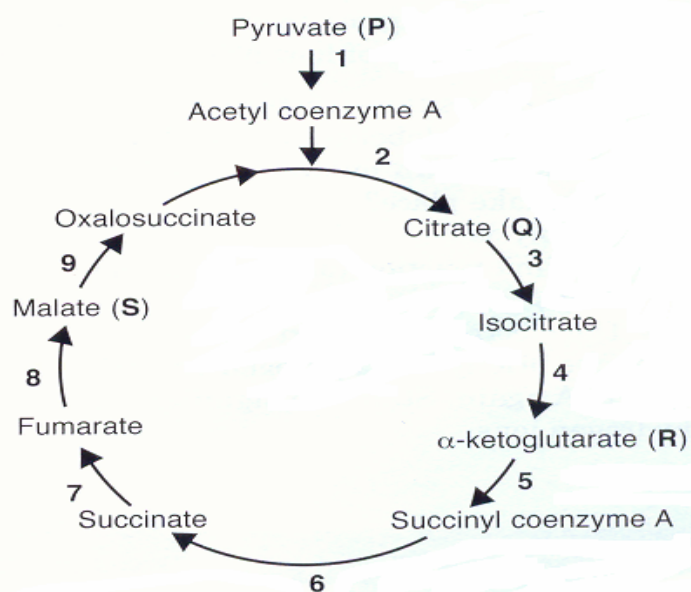
- A Green wavelengths are absorbed and red and blue wavelengths are transmitted.
- B Red and blue wavelengths are absorbed and green wavelengths are transmitted.**
- C Blue wavelengths are absorbed and green and red wavelengths are transmitted.
- D Green and red wavelengths are absorbed and blue wavelengths are transmitted.

24. The diagram below shows the stages in the Calvin cycle. Which of the stages use ATP?



- A Stage 1 and Stage 3**
- B Stage 2 and Stage 4
- C Stage 1 and Stage 2
- D Stage 3 and Stage 4

25.



At which stages are hydrogen atoms transferred to NAD^+ ?

- A 1, 4, 5, and 7
- B 1, 4, 5, and 9**
- C 2, 4, 5, and 7
- D 4, 5, 7, and 9

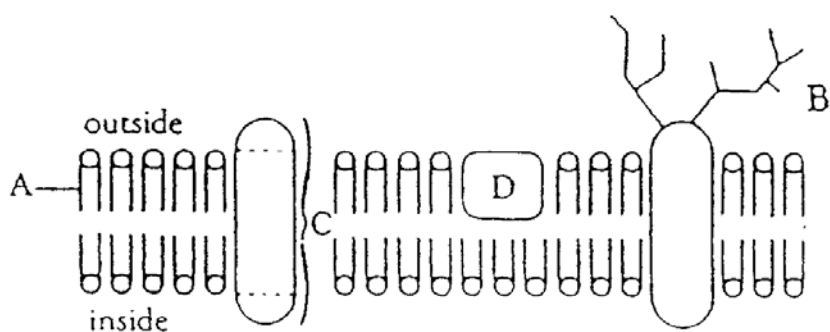
26. What is the role of oxygen in cell respiration?

- A to act as the final hydrogen acceptor in the electron transport chain**
- B to combine with acetyl coenzyme A to form citrate in the Krebs cycle
- C to combine with carbon from carbohydrates to form carbon dioxide
- D to oxidise reduced NAD in oxidative phosphorylation

27. What products are formed when anaerobic respiration takes place in muscles and yeast cells.

- | | Muscles | Yeast Cells |
|----------|--|--|
| A | pyruvate, NAD^+ , ATP | ethanol, NAD^+ , ATP |
| B | ethanol, NADH, ADP, CO_2 | lactate, NADH, ADP, CO_2 |
| C | lactate, NAD^+ , ATP, CO_2 | ethanol, NAD^+ , ATP |
| D | lactate, NAD^+, ATP | ethanol, NAD^+, ATP, CO_2 |

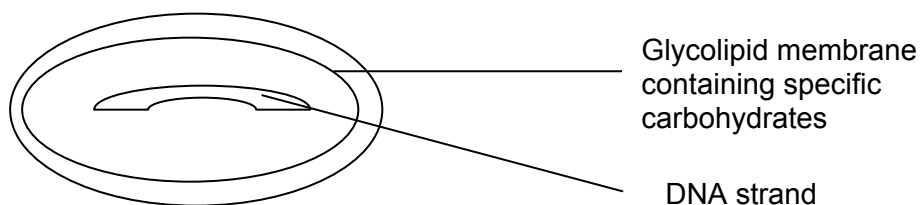
28. Which structure(s) contain(s) the presence of a peptide bond?



- A D only
- B A and B
- C C and D**
- D B, C and D

29. Gene therapy is a way of treating genetic disease by introducing a piece of DNA into the cells of an affected individual. Liposomes can be used for gene therapy as they target the cells affected by a genetic disease.

The diagram shows the structure of a liposome.



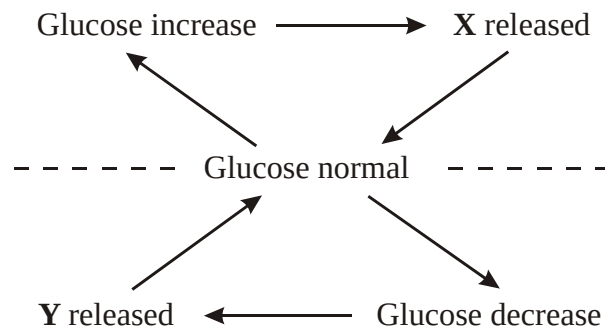
Which feature of a cell membrane allows the liposome to target cells affected by a genetic disease?

- A carrier molecules
- B phosphate groups
- C protein channels
- D receptor molecules**

30. Protein phosphorylation is usually involved with all of the following **except**

- A activation of G-protein linked receptors.**
- B activation of protein kinase molecules.
- C enzyme activation.
- D regulation of transcription by extracellular ligands.

31. The diagram shows how the body regulates glucose levels in the blood.



What is Y?

- A amylase
- B insulin
- C glucagon
- D glycogen

32. What is the role of active transport in the transmission of nerve impulses by neurones?

- A Propagates an action potential by pumping sodium ions across the membrane out of the neurone.
- B Propagates an action potential by pumping sodium ions across the membrane into the neurone.
- C Initiates the action potential needed for the transmission of an impulse by pumping calcium ions out of the endoplasmic reticulum.
- D Establishes the resting potential needed for the transmission of an impulse by pumping sodium and potassium ions across the membrane

33. In a nerve impulse, what happens at the site following the highest point of the action potential?

- A Voltage-gated sodium ion channels open and Na^+ is pumped in.
- B Voltage-gated sodium ion channels open and Na^+ diffuses out.
- C Voltage-gated potassium ion channels open and K^+ is pumped out.
- D Voltage-gated potassium ion channels open and K^+ diffuses out.

34. A drug might stimulate the nervous system if it

- A increases the efflux of K^+ from axoplasm that cause the hyperpolarisation of the neurons.
- B increases the sensitivity of the post-synaptic membrane to sodium.
- C increases the sensitivity of the pre-synaptic membrane to sodium.
- D makes the membrane permanently impermeable to sodium.

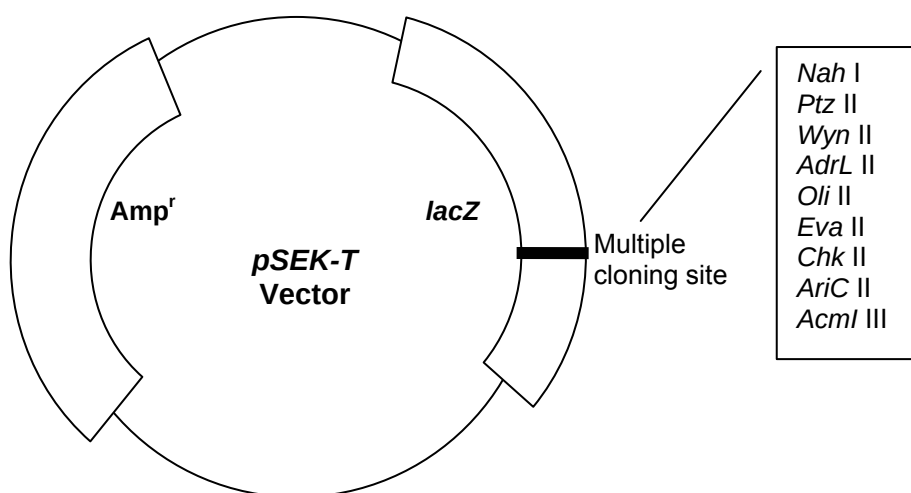
35. Why has evolution resulted in the appearance of antibiotic resistant bacteria?

- A Bacteria learn the ability to neutralize the effects of antibiotic and they pass this trait to the next generation.
- B Bacteria modify their metabolism to cope with the presence of antibiotics.
- C Bacteria resistant to the antibiotic survive and pass on this characteristic to their next generation.
- D Bacteria develop resistant due to the incomplete course of antibiotic.

36. Which of the following situation would evolution be fastest for an interbreeding population?

	Migration	Selection pressure	Variation due to mutation
A	High rate	High	High
B	High rate	Low	High
C	Low rate	High	High
D	Low rate	Low	Low

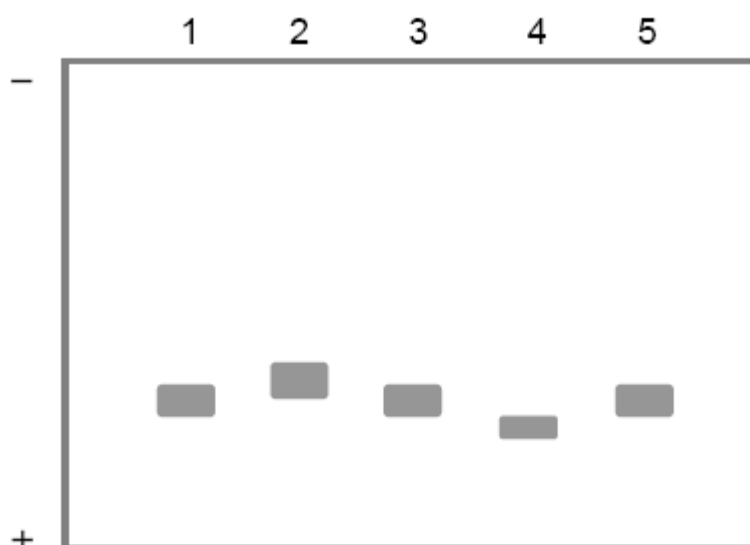
37. As part of the procedure to produce recombinant proteins in *E. coli*, you are asked to insert the gene encoding for the D-ONG protein into the pSEK-T vector. The restriction sites and selectable markers on the vector are shown below.



If the gene for D-ONG protein were to be inserted into the multiple cloning site, what should be added to the agar plate in order to screen for recombinant clones and how would the recombinant clones appear?

	Chemicals to be added		Colour of colonies
A	Ampiciliin	X-gal	Blue
B	β-galactosidase	X-gal	Blue
C	Ampiciliin	X-gal	White
D	β-galactosidase	lactose	White

38. The diagram shows the results of SDS polyacrylamide gel electrophoresis (PAGE) of the same protein extracted from five different individuals.



Which of the following correctly explains why the proteins extracted from individuals 2 and 4 did not travel the same distance on the gel as those extracted from individuals 1, 3 and 5?

	individual 2	individual 4
A	The protein has an additional positively charged amino acid.	A non-charged amino acid in the protein has been replaced by a negatively charged amino acid.
B	The protein has an additional amino acid.	An amino acid in the protein has been deleted.
C	The protein has an additional amino acid.	A non-charged amino acid in the protein has been replaced by a negatively charged amino acid.
D	The protein has an additional positively charged amino acid.	An amino acid in the protein has been deleted.

39. Which of the following statement about stem cells is correct?

- A Stem cells are specialised cells that have yet to express the genes and proteins characteristic of their differentiated state, and do so when needed for repair of tissues and organs.
- B Stem cells are fully differentiated cells that reside under the surface of epithelia, in position to take over the function of the tissue when the overlying cells become damaged or worn out.
- C Stem cells are undifferentiated embryonic cells that persist in the adult, and can give rise to all the cell types in the body.
- D Stem cells are undifferentiated cells that divide asymmetrically, giving rise to one daughter that remains a stem cell, and one daughter that will differentiate to replace damaged and worn out cells in the adult tissue or organ.

40. One type of genetically modified corn has

- A gene for the production of Bt toxin which protects the plant against a specific insect;
- An 'amp'-gene which was introduced in the plant together with the Bt toxin gene. This gene gives resistance to the antibiotic ampicillin.

There is concern that the 'amp' gene may transfer to enterobacteria in the human intestine during nucleic acid digestion making treatment with ampicillin ineffective for diseases caused by enterobacteria.

Which statement explains why the transfer of this gene from the plant to bacteria in the human intestine is unlikely?

- A** An origin of replication and appropriate prokaryotic promoters are required for the 'amp' gene to be expressed.
- B** Bacteria took up DNA released during digestion of the plants by human nuclease enzymes, without the presence of a vector.
- C** 50% of the enterobacteria isolated from humans are 'amp' resistant.
- D** All plant DNA is digested and destroyed in the human intestine during digestion of plant cells by human nuclease enzymes.